

Muralidhar Kolimali

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Professional Summary

Recent Master's graduate in Data Science with professional experience in AI-powered applications, backend engineering, data analytics, and large-scale data processing. Skilled in building scalable AI systems using Python, FastAPI, PostgreSQL, OpenAI GPT models, Retrieval-Augmented Generation (RAG), and REST APIs. Experienced in developing production-oriented AI workflows, conversational AI systems, backend automation, and data pipelines. Strong understanding of machine learning, NLP concepts, API integrations, cloud-ready architectures, and enterprise application development.

Technical Skills

Programming Languages	Python, SQL, PL/SQL, TypeScript/JavaScript (Basics), Pro*C, Shell Scripting
AI & Machine Learning	Machine Learning, NLP, RAG Pipelines, Prompt Engineering, OpenAI GPT-4o-mini, Scikit-learn, YOLOv3, Q-Learning
Backend & APIs	FastAPI, REST APIs, Webhooks, Twilio API, API Integrations, OAuth Concepts
Databases & Data Engineering	PostgreSQL, SQL Optimization, ETL Concepts, Data Pipelines, Data Validation
Cloud & DevOps	Docker, Docker Compose, Git/GitHub, CI/CD Concepts, Cloud-Native Deployment Concepts, AWS Familiarity
Tools & Frameworks	FAISS, Sentence Transformers, Pandas, NumPy, OpenCV, Power BI
Certifications	NVIDIA LLM & Chatbot Development Workshop

Professional Experience

Data Analytics & Data Engineering Intern 2025 – Present
Eastern Metal Works (EMW), Connecticut, USA

- Worked on operational analytics and dashboard development projects involving ETL workflows, Power BI reporting, and business data integration.
- Assisted in integrating and analyzing enterprise operational datasets to support reporting and decision-making workflows.
- Collaborated on backend data preparation, data transformation, and reporting automation processes.
- Supported dashboard enhancements, manual tagging systems, and operational data filtering workflows for business analytics use cases.

Software Engineer June 2022 – July 2024
Virtusa Consulting Services (Client: Citi India), Chennai, India

- Worked on enterprise banking migration projects involving large-scale retail banking datasets and backend operational systems.
- Developed and optimized PL/SQL queries using indexing, analytical functions, and optimizer hints to improve data processing performance.
- Automated backend operational workflows using Pro*C and Shell scripting, improving process efficiency and production reliability.
- Supported UNIX-based production systems, backend monitoring, procedural testing, and root-cause analysis activities.
- Collaborated with cross-functional teams for backend integration, enterprise support, and large-scale data validation workflows.

Projects

AI Voice Assistant for Government Services

- Developed an AI-powered conversational voice assistant using FastAPI, OpenAI GPT-4o-mini, Twilio, PostgreSQL, and Retrieval-Augmented Generation (RAG).
- Built intelligent conversational workflows for healthcare, transportation, caregiving, and support service applications.

- Designed scalable REST APIs and backend services to process real-time voice interactions and contextual AI responses.
- Implemented vector search pipelines using FAISS and Sentence Transformers for intelligent document retrieval.
- Containerized backend services using Docker and Docker Compose for scalable deployment workflows.
- Developed backend pipelines for storing user interactions, call logs, and transcripts using PostgreSQL.
- Applied prompt engineering and conversational AI techniques to improve contextual response generation and user interaction quality.

Vehicle Detection using YOLOv3

- Developed a computer vision pipeline for vehicle detection and classification using YOLOv3 and OpenCV.
- Implemented frame-by-frame object detection workflows and optimized inference handling for traffic monitoring applications.

MindMaze – Intelligent Puzzle Game

- Built an adaptive puzzle game using machine learning-driven difficulty adjustments and Q-Learning concepts.
- Developed a real-time hint recommendation system to improve user interaction and gameplay experience.

Workshops & Certifications

- NVIDIA LLM & Chatbot Development Workshop (2025) – Hands-on training in prompt engineering, LLM fine-tuning, conversational AI, and AI deployment workflows.
- Participated in advanced university-hosted AI workshops focused on scalable AI application development and model deployment concepts.

Education

Master of Science in Data Science	May 2026
University of New Haven, Connecticut	GPA: 3.5
Bachelor of Technology in Electronics and Communication Engineering	
Raghu Institute of Technology, India	GPA: 8.32